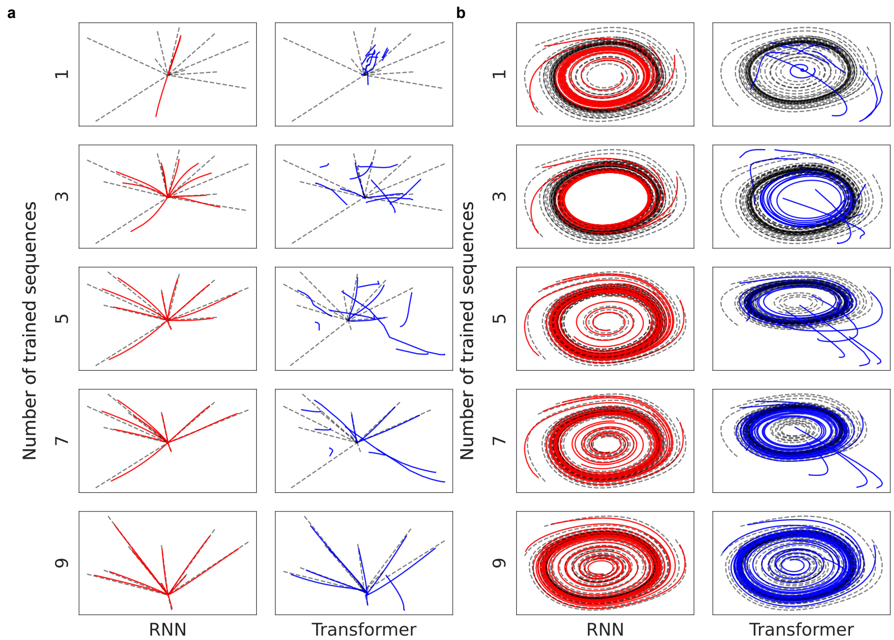

Figure 2: **Qualitative evaluation of model performance. a, b,** The outputs of trained models using point attractor and cyclic attractor dynamics, respectively, employ varying numbers of sequences. Colored lines represent the outputs, while the true trajectory corresponding to the provided initial position is depicted using dot lines.

cyclic attractors. We systematically varied the number of training sequences to examine the quantity of training data necessary to extract essential structure underlying a dataset (see Methods; **trajectories of attractor dynamics**). Given a random initial position denoted by xy coordinates, the models were required to predict the position one step ahead in an autoregressive manner for T time steps. For performance evaluation, the Dynamic Time Warping (DTW) method has been employed (see Methods; **evaluation settings**).

These experimental results showed that the RNN successfully acquired attractor dynamics using significantly fewer data than the Transformer model. Interestingly, as illustrated in Figure 1, the RNN converged to almost the correct trajectory with only 2-4 training datasets, whereas the Transformer model required approximately 25-50 datasets to achieve similar performance (for more detail, see Appendix; Table 1 and 2). In Figure 2, a qualitative evaluation reveals that the RNN already achieved a highly satisfactory representation with just three sequences. Notably, in the realm of cyclic attractors, the RNN model produced output that converged upon a limit cycle, even when trained on a single sequence. Conversely, the Transformer model struggled to reproduce attractors under the same initial conditions, despite being trained with nine sequences. It is important to note that we conducted the same experiments involving varying dropout rates to address concerns related to the Transformer's susceptibility to overfitting, and the results remained consistent (see Appendix; **Model variations to avoid overfitting**). Furthermore, we extended our comparative analysis to address the inherently more complex task of human hand-drawn pattern learning (see Appendix; **Human hand-drawn pattern learning**).

Discussion

This study investigated the difference in generalization in learning (GIL) between RNN and Transformer. Our results revealed that the RNN model has the capacity to extrapolate fundamental dynamic structure of attractors from even a limited number of data, in contrast to the Transformer model, which requires considerably more data for the same task. These results support our hypothesis that generalization by Transformer is considerably weaker when working with limited numbers of exemplars.

Compared to recurrent models, the deficiency of feed-forward models, which are utilized in the Transformer model, has been reported in various tasks [14, 15, 16]. This shortfall in feed-forward models is often attributed to their inability to acquire "recurrent" inductive bias [10, 12]. From a different perspective, Elman et al. [17, 18, 19, 20] have demonstrated that distributed representation in neural networks can yield a global structure that accounts for both learned and unlearned patterns, while local representation [21, 22, 23] is inferior in its capacity to capture global data structure [24, 25]. Given reports, we attribute the dissimilarity of the GILs observed in this study to the ability to acquire "recurrent" inductive bias inherent to shared weights.

Given that the artificial intelligence community has overwhelmingly favored scaling as the primary method to improve performance, the study of GIL with limited exemplars has been somewhat overlooked [26, 9]. Nevertheless, this study reveals substantial disparities in GIL when working with limited data. Our findings have important implications and underscore potential relevance for architecture selection in practical applications where data scarcity is a common constraint, such as medicine, finance, and robotics. We acknowledge there are several limitations to this study, notably that our experiment uses only the vanilla Transformer-based model. Hence, further investigations employing recently proposed models will be conducted [12, 27, 28], which employs the "recurrent" inductive bias. We hope that this work will motivate others to further investigate GIL with limited exemplars and that these insights will be useful in developing the next generation of neural networks.

METHODS

Trajectories of attractor dynamics. Since our primary focus was to evaluate the capacity of GIL in the models, two relatively simple types of attractor dynamics, i.e., point and cyclic attractors, were used as training trajectories. Each trajectory was initiated from random initial positions in the xy-coordinate range of [-3.0, 3.0] and transited for T time steps. Point attractors were generated utilizing equations 1 with the parameter α set to -1.0 and T set to 100. This equation lets the point at a random initial position move linearly closer